GIDFINE LOAN PLATFORM

PHASE 4A POST-IMPLEMENTATION AUDIT REPORT

Independent Read-Only Review | Branch: development | HEAD: 931411f

Audit Date: October 21, 2026	**Target Branch:** development (HEAD @ 931411f)
Audit Mode: Strictly Read-Only (0 edits, 0 commits)	**main Branch:** Untouched (at f7ed8d1)
Investigation Focus: Schema indexes & Idempotency	**Next Step:** Stopped before Phase 4B pending review

1. Executive Summary & Git Integrity

A rigorous, read-only post-implementation audit was conducted on the Phase 4A codebase. The active branch is confirmed as development at commit 931411f (5 commits ahead of origin, 0 pushed). Branch main is confirmed completely untouched at f7ed8d1. Zero files were modified, committed, pushed, or deployed during this audit review.

2. Ledger Schema Consistency & Discrepancy Investigation

Investigation of ledgerEntries.by_agency_and_idempotency vs idempotencyKey:

• **Actual Code:** In frontend/convex/schema.ts at line 215, idempotencyKey: v.optional(v.string()) is explicitly declared on the ledgerEntries table. The compound index at line 222 (by_agency_and_idempotency: ['agencyId', 'idempotencyKey']) references a valid field that exists in the database schema.

• **Discrepancy Source:** The summary table in the Phase 4A implementation PDF omitted idempotencyKey from the bulleted field list of ledgerEntries while displaying the index in the right-hand column, causing documentation confusion.

• **Optional Field Consequence:** Because idempotencyKey is optional on ledgerEntries, documents without an idempotency key will index with undefined. In Phase 4B, queries must explicitly handle undefined values.

3. Idempotency & Concurrency Evaluation

• **Convex Database Limits:** In Convex v1.47.0, secondary table indexes do not provide native unique constraints.

• **ACID OCC Guarantees:** Convex mutations run with strict serializable ACID isolation using Optimistic Concurrency Control (OCC). When an index query (.withIndex('by_agency_and_idempotency', ...)) is performed within a mutation, Convex records the read interval. A concurrent insert on that key invalidates the read set, triggering an automatic OCC abort and retry.

• **Phase 4B Requirement:** To guard against caller payload drift on retry, the mutation must verify that matching idempotency keys also match the request parameters (amount, application), throwing an error if parameters differ.

4. Maker-Checker Security Audit

• **Current Enforcement:** Maker-Checker segregation (makerId !== checkerId) is currently defined in helper logic (validateMakerCheckerSegregation in ledgerTypes.ts) and tested in ledgerSchema.test.ts, but is **NOT yet enforced in live backend mutations**, because Phase 4A prohibited implementing business logic mutations.

• **Required Phase 4B Guard:** In the upcoming approval mutation, identity must be extracted strictly from ctx.auth.getUserIdentity() (never client input), and the mutation must assert disbursementRequest.makerId !== currentUser._id.

5. Ledger Accounting Model & Open Decisions

• **What is Supported:** Append-only immutable records, integer pesewas precision, GHS currency restriction, compensating reversals (Phase 4C), and multi-tenant isolation via agencyId.
• **What is NOT Supported:** No double-entry general ledger (no debit/credit balancing legs, no chart of accounts), no materialized running balances, and no persistent repayment schedule.
• **Decisions for Matthias:** 1) Balance calculation (dynamic aggregation vs cached field); 2) Repayment schedule storage model; 3) Waterfall allocation priority; 4) Reversal dual-authorization requirements.

6. Reproducible Test Execution Results

Test Suite	Command	Exit Code	Results
TypeScript Compiler	`npx tsc --noEmit`	0	PASSED (0 errors)
Node Test Runner (All)	`node --test tests/*.test.ts`	0	PASSED (42 / 42 tests)
Phase 4A Tests	`node --test tests/ledgerSchema.test.ts`	0	PASSED (12 / 12 tests)
Python Regression	`pytest tests/ -q`	0	PASSED (6 / 6 tests)

7. Classified Findings Matrix

ID	Sev	Location	Summary & Finding Details
F-01	MEDIUM	PDF Summary Table	**Documentation Discrepancy:** Phase 4A PDF omitted idempotencyKey from the bullet list of ledgerEntries fields, despite existing in schema.ts:215.
F-02	MEDIUM	`schema.ts:215`	**Optional Key in Compound Index:** idempotencyKey is optional on ledgerEntries, indexing as undefined when omitted.
F-03	LOW	`ledgerTypes.ts:173`	**Unwired Maker-Checker Guard:** Helper exists but is not yet wired to a mutation (deferred to Phase 4B per scope boundary).
F-04	LOW	`ledgerSchema.test.ts`	**Idempotency Test Gap:** Test covers identical payload retries but does not assert payload mismatch errors on key reuse.

8. Final Audit Status

Status: STOPPED. Phase 4B has not been started. The audit review is complete with zero changes applied.